

Combinatorics – 2

Recurrences and generating functions

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Theoretical minimum. A generating function is not a function: it is a way of writing a whole sequence as one object, so that operations on sequences become algebra. Convergence is irrelevant here — the analytic side, radii and growth of coefficients, is Functional sequences, series and spaces – 3.

Definitions. The ring $\mathbb{F}[[x]]$ of *formal power series* consists of all expressions $\sum_{n \geq 0} a_n x^n$ with the obvious addition and with multiplication $(\sum a_n x^n)(\sum b_n x^n) = \sum_n (\sum_k a_k b_{n-k}) x^n$: the coefficients of a product are the convolution of the coefficients. A series is invertible exactly when $a_0 \neq 0$; in particular $(1 - x)^{-1} = 1 + x + x^2 + \dots$ is an identity in $\mathbb{F}[[x]]$ requiring no convergence. Composition $F(G(x))$ is defined whenever $G(0) = 0$. The *ordinary* generating function of (a_n) is $\sum a_n x^n$, the *exponential* one is $\sum a_n x^n / n!$.

Theorem 1 (linear recurrences). Let $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ for $n \geq k$. Then $A(x) = \sum a_n x^n = \frac{P(x)}{Q(x)}$ with $Q(x) = 1 - c_1 x - \dots - c_k x^k$ and $\deg P < k$; conversely every rational series with $Q(0) \neq 0$ satisfies such a recurrence. Factoring $Q(x) = \prod (1 - \lambda_i x)^{m_i}$ and expanding into partial fractions gives

$$a_n = \sum_i p_i(n) \lambda_i^n, \quad \deg p_i < m_i,$$

where the λ_i are the roots of the *characteristic polynomial* $x^k - c_1 x^{k-1} - \dots - c_k$. Multiple roots produce the polynomial factors, which is exactly where a naive guess fails.

Theorem 2 (the dictionary). Choosing an object from a disjoint union adds generating functions; building a pair adds exponents, so it multiplies them; a sequence of arbitrarily many pieces each counted by F with $F(0) = 0$ is counted by $1/(1 - F)$. Hence: compositions of n into parts from S

have generating function $1/(1 - \sum_{s \in S} x^s)$; partitions into parts from S have $\prod_{s \in S} (1 - x^s)^{-1}$; labelled structures multiply as exponential generating functions, and a set of components counted by F gives e^F .

Theorem 3 (transfer matrix). If the objects of size n are the walks of length n in a finite directed graph with adjacency matrix M , then their number is the corresponding entry of M^n , the generating function is a rational function with denominator $\det(I - xM)$, and the growth rate is the spectral radius of M — which for a non-negative matrix is the Perron root (Linear algebra – 11). This is how one counts words avoiding a pattern and tilings of a strip.

1. Prove Theorem 1 in both directions, including the case of multiple roots. Then solve $a_n = 4a_{n-1} - 4a_{n-2}$, $a_0 = 1$, $a_1 = 3$, and prove that a sequence of integers satisfying a linear recurrence with integer coefficients and $|\lambda_i| < 1$ for all roots is eventually zero.
2. Work in $\mathbb{F}[[x]]$ without any convergence: prove that $\sum a_n x^n$ is invertible if and only if $a_0 \neq 0$, compute the inverse of $1 - x - x^2$, and prove the two dictionary rules of Theorem 2. Then find the generating function for the number of compositions of n into parts 1 and 2 and identify the sequence.
3. Let $x_0 = 1$ and $x_{n+1} = 3x_n + \lfloor x_n \sqrt{5} \rfloor$ for $n \geq 0$, so that $x_1 = 5$, $x_2 = 26$, $x_3 = 136$, $x_4 = 712$. Find a closed-form expression for x_n . (Putnam)
4. Suppose X is a random variable taking only non-negative integer values with $\mathbb{E} X = 1$, $\mathbb{E} X^2 = 2$ and $\mathbb{E} X^3 = 5$. Determine the smallest possible value of $\Pr(X = 0)$. (Putnam)
5. Transfer matrices: count the binary words of length n with no two consecutive ones, and the domino tilings of a $2 \times n$ and of a $3 \times n$ rectangle; in each case write the rational generating function, identify

the characteristic polynomial and compute the exponential growth rate as a spectral radius.